

Task 5 - Air Quality Monitoring System

Project Title

Air Quality Monitoring System using MQ-135 Sensor and Arduino

Objective

To design a system that detects air pollution levels using a gas sensor, displays readings, and stores results for monitoring.

Duration

8 Hours

Project Complexity

Medium

Learning Outcomes

- Understand gas sensor interfacing.
- Learn air quality measurement basics.
- Display and log environmental data.
- Build pollution monitoring solutions.

Components Required

- Arduino Uno
- MQ-135 Air Quality Sensor
- 16x2 LCD Display (I2C optional)
- Buzzer (optional)
- SD Card Module / Storage
- Jumper Wires
- Breadboard
- USB Cable / Power Supply

Working Principle

1. MQ-135 senses gases such as smoke, NH₃, CO₂ and pollutants.
2. Arduino reads analog sensor value.
3. Value is converted into air quality index level.
4. Reading is displayed on LCD/Serial Monitor.

5. Data can be stored for later analysis.
6. Alert activates when pollution exceeds threshold.

Circuit Connections

- MQ-135 AO -> A0
- MQ-135 VCC -> 5V
- MQ-135 GND -> GND
- LCD SDA -> A4
- LCD SCL -> A5
- LCD VCC -> 5V
- LCD GND -> GND
- Buzzer + -> D8
- Buzzer - -> GND

Arduino Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);

int gasPin = A0;
int buzzer = 8;
int value = 0;

void setup() {
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
}

void loop() {
  value = analogRead(gasPin);

  Serial.print("Air Quality Value: ");
  Serial.println(value);

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Air Value:");
  lcd.setCursor(0,1);
  lcd.print(value);

  if(value > 400) {
    digitalWrite(buzzer, HIGH);
```

```
} else {  
    digitalWrite(buzzer, LOW);  
}  
  
delay(2000);  
}
```

Sample Output

```
Air Quality Value: 215  (Good)  
Air Quality Value: 480  (Poor - Alert)
```

Real-World Applications

- Smart city pollution monitoring
- Indoor office air monitoring
- Factory safety systems
- Home health monitoring

Future Enhancements

- IoT cloud dashboard
- Mobile notifications
- PM2.5 sensor integration
- GPS-based pollution mapping

Conclusion

This project helps monitor harmful gases and pollution levels in real time. It is valuable for health, safety, and smart city applications.